Year 11 Set 2 · Higher · grade 5–6 · Lesson 5

Arcs, sectors & volume

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Today I will learn to...

Goals

1. Find arc lengths and sector areas

2. Find the volume of cylinders, cones and spheres

Do now: retrieval practice

4 mins

1. Area of a circle with radius 3, in terms of π

   9π

2. Circumference of a circle with diameter 10 (1 dp)

   31.4

3. Simplify 90/360

   1/4

4. Work out 6 × 4.5

   27

Objective 1

Find arc lengths and sector areas

Key idea

Learn

!

A sector is the fraction θ/360 of a circle. Arc length = θ/360 × 2πr. Sector area = θ/360 × πr².

Objective 1: Find arc lengths and sector areas

I do: example 1

I do

Arc length: radius 6 cm, angle 60°

1. $60/360 = 1/6$
2. $1/6 \times 2 \times \pi \times 6 = 2\pi$
3. $= 6.28$ cm

I do: example 2

I do

Sector area: radius 10 cm, angle 72°

1. $72/360 = 1/5$
2. $1/5 \times \pi \times 100 = 20\pi$
3. $= 62.8\ cm^2$

I do: example 3

I do

Perimeter of a sector: radius 5 cm, angle 120°

1. Arc = $1/3 \times 10\pi = 10.47$ cm
2. Add the two radii: $10.47 + 5 + 5$
3. $= 20.5$ cm

Mini whiteboards

We do

1. Sector area, r = 4, angle 90° (1 dp)

Answer: $12.6\ cm^2$

2. Arc length, r = 9, angle 40° (1 dp)

Answer: 6.3 cm

3. Arc length, r = 6, angle 60°, in terms of π

Answer: 2π cm

4. Sector area, r = 10, angle 72°, in terms of π

Answer: $20\pi\ cm^2$

You do: choose your challenge

You do

Bronze

Area of a quarter circle with radius 10 (1 dp)

Silver

Arc length: radius 12, angle 150° (1 dp)

Gold

Perimeter of a sector: radius 7, angle 80° (1 dp)

Platinum

A sector has arc length 4π cm and radius 9 cm. Find its angle and its area in terms of π.

You do: answers

Check

Bronze	78.5
Silver	$10\pi = 31.4$
Gold	9.77 + 14 = 23.8
Platinum	80°, area 18π cm^2

If you finish early

Extra

Bronze

Find the area of a sector with radius 6 cm and angle 120°, in terms of π.

12π cm²

Silver

Find the arc length of a sector with radius 15 cm and angle 72° (1 dp).

$6\pi = 18.8$ cm

Gold

Find the perimeter of a sector with radius 10 cm and angle 54° (1 dp).

$3\pi + 20 = 29.4$ cm

Platinum

A sector has area 15π cm² and radius 6 cm. Find its angle.

$\theta/360 \times 36\pi = 15\pi$, so $\theta = 150°$

Exam question 1

2 marks

A sector has radius 8 cm and angle 135°. Find its area to 1 dp.

Space for working

Mark scheme: $135/360 \times \pi \times 64 = 75.4\ \text{cm}^2$

Exam question 2

3 marks

Find the perimeter of the same sector to 1 dp.

Space for working

Mark scheme: Arc = 135/360 × 16π = 18.85. 18.85 + 16 = 34.8 cm

Objective 1 achieved

Find arc lengths and sector areas



Objective 2

Find the volume of cylinders, cones and spheres

✓ Achieved: Find arc lengths and sector areas

Key idea

Learn

!

Cylinder = $\pi r^2 h$. Cone = $\frac{1}{3}\pi r^2 h$. Sphere = $4/3\ \pi r^3$. The cone and sphere formulas are given on the exam paper.

Objective 2: Find the volume of cylinders, cones and spheres

I do: example 1

I do

Cylinder: radius 3 cm, height 10 cm

1. Volume of a cylinder = $\pi \times r^2 \times h$
2. $r^2 = 3^2 = 9$
3. $\pi \times 9 \times 10 = 90\pi$
4. $= 282.7\ cm^3$ (1 dp)

I do: example 2

I do

Cone: radius 4 cm, height 9 cm

1. Volume of a cone = $\frac{1}{3} \times \pi \times r^2 \times h$
2. $r^2 = 4^2 = 16$
3. $\frac{1}{3} \times \pi \times 16 \times 9 = 48\pi$
4. $= 150.8\ cm^3$ (1 dp)

I do: example 3

I do

Sphere: radius 6 cm

1. Volume of a sphere = $4/3 \times \pi \times r^3$
2. $r^3 = 6^3 = 216$
3. $4/3 \times \pi \times 216 = 288\pi$
4. $= 904.8\ cm^3$ (1 dp)

Mini whiteboards

We do

1. Cylinder r = 2, h = 5, in terms of π

Answer: 20π

2. Hemisphere r = 3, in terms of π

Answer: 18π

3. Cone r = 3, h = 4, in terms of π

Answer: 12π

4. Sphere r = 6, in terms of π

Answer: 288π

You do: choose your challenge

You do

Bronze

Cylinder r = 5, h = 8, in terms of π

Silver

Cone r = 6, h = 10 (1 dp)

Gold

Hemisphere radius 9 cm (1 dp)

Platinum

A cylinder of radius 4 cm holds water. A sphere of radius 3 cm is dropped in and sinks. How far does the water rise?

You do: answers

Check

Bronze

200π

Silver

$120\pi = 377.0$

Gold

$486\pi = 1\,526.8\ \text{cm}^3$

Platinum

$36\pi \div 16\pi = 2.25\ \text{cm}$

If you finish early

Extra

Bronze

Cylinder radius 3 cm, height 10 cm. Volume in terms of π.

90π cm³

Silver

Cone radius 4 cm, height 9 cm. Volume (1 dp).

$48\pi = 150.8$ cm³

Gold

Sphere diameter 10 cm. Volume (1 dp).

$500\pi/3 = 523.6$ cm³

Platinum

A cone and a cylinder both have radius 3 cm. The cylinder is 8 cm tall and the cone has the same volume. How tall is the cone?

$72\pi = 3\pi h$, so $h = 24$ cm

Exam question 1

2 marks

A cylinder has diameter 10 cm and height 12 cm. Find its volume to 1 dp.

Space for working

Mark scheme: $\pi \times 25 \times 12 = 942.5\ \text{cm}^3$

Exam question 2

3 marks

A cone has radius 5 cm and volume 200 cm^3. Find its height to 1 dp.

Space for working

Mark scheme: 200 = ⅓ × π × 25 × h, h = 600 ÷ 25π = 7.6 cm

Objective 2 achieved

Find the volume of cylinders, cones and spheres



Real life

Apply

A cylindrical water tank has radius 0.5 m and height 1.2 m. How many litres does it hold? (1 m^3 = 1 000 litres)

$\pi \times 0.25 \times 1.2 = 0.942\ m^3 = 942$ litres

Extension: stretch yourself

Stretch

1. A sector has area 30π cm^2 and radius 10 cm. Find its angle.

$\theta/360 \times 100\pi = 30\pi$, $\theta = 108°$

2. A sphere has volume 36π cm^3. Find its radius.

$4/3\ r^3 = 36$, $r^3 = 27$, $r = 3$ cm

Exit ticket: exam questions

Exit

1. Find the area of a sector with radius 6 cm and angle 50° (1 dp). **(2 marks)**

2. Find the volume of a cylinder with radius 4 cm and height 7 cm (1 dp). **(2 marks)**

3. Find the volume of a sphere with diameter 12 cm, in terms of π. **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 $5\pi = 15.7\ cm^2$ [2]

Q2 $112\pi = 351.9\ cm^3$ [2]

Q3 $288\pi\ cm^3$ [2]

Total: ____ / 6

Today I achieved

- ✔ **Find arc lengths and sector areas**
- ✔ **Find the volume of cylinders, cones and spheres**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Find the arc length of a sector with radius 15 cm and angle 48° (1 dp). [2]

2. Find the area of a sector with radius 9 cm and angle 200° (1 dp). [2]

3. A cone has radius 3 cm and height 8 cm. Find its volume (1 dp). [2]

4. A cylinder holds 500 cm^3 and has radius 5 cm. Find its height (1 dp). [3]

5. A solid is a cylinder (radius 2 cm, height 6 cm) with a hemisphere of radius 2 cm on top. Find its volume (1 dp). [3]

Extra practice: Maths Genie → Edexcel question bank → Arcs and Sectors · Volume